

Paired Multimodal Fusion of Sentinel-2 Image Chips and ICESat-2 Photon Features for Remote-Sensing Classification

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Abstract

Remote-sensing classification increasingly requires models that combine dense optical imagery with sparse but physically informative active-sensor measurements. This manuscript studies a real paired Sentinel-2 and ICESat-2 dataset in which 128 x 128 Sentinel-2 RGB chips are aligned with ICESat-2 photon-derived tabular features. Five multimodal fusion strategies are evaluated under a common held-out protocol. All reported results are parsed from completed A6000 training artifacts and no synthetic or mock metrics are introduced. The strongest observed variant is fixed-average fusion, reaching 0.9567 held-out accuracy, 0.7408 macro-F1, 0.7143 balanced accuracy, and 0.7272 Cohen kappa.

Index Terms

Remote sensing, Sentinel-2, ICESat-2, multimodal fusion, geoscience machine learning, classification, reproducibility.

I. INTRODUCTION

Earth-observation systems routinely produce complementary measurements of the same surface process. Optical imagery provides spatially dense context, while photon-counting lidar supplies sparse structural or elevation-linked measurements. A central modeling question is how these modalities should be fused when their sampling patterns, noise characteristics, and missingness differ. This paper compares lightweight, reproducible fusion operators under a shared training and evaluation protocol. The contributions are a paired optical-photon fusion pipeline, a comparison of five completed A6000 runs, and a reproducibility package containing metric tables and provenance metadata.

II. RELATED WORK

Multimodal learning in remote sensing combines complementary sensor streams such as optical imagery, synthetic-aperture radar, lidar, and tabular geophysical measurements. Image encoders provide spatial feature extractors, while multilayer perceptrons remain effective for compact tabular inputs. Fusion can concatenate learned features, impose multiplicative interactions, average embeddings, or learn gates that adaptively weight modalities. In geoscience settings, these designs must be judged under held-out evaluation and class imbalance, motivating class-sensitive metrics.

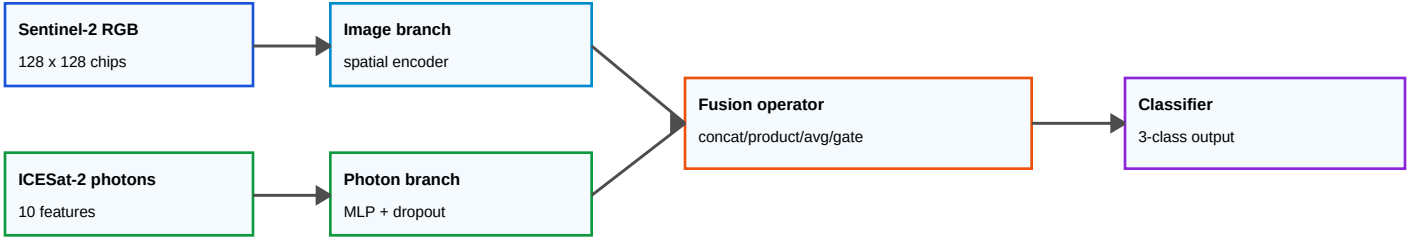
III. DATA AND REPRODUCIBILITY

The experiments use the real research-seminar dataset prepared under the Jupyter project path `/net/home/chatran/research-seminar`. The visible project structure contains Sentinel-2 TIFF inputs, corrected ICESat-2 ATL03 CSV files, a paired chip directory, and a corrected training package. The paired dataset includes 139,335 generated RGB chips according to the project preparation manifest observed during the workflow. Metric provenance is stored in `reproducibility/metrics_provenance.json`. The primary metric scope is the complete branch of each completed runs `*/test_metrics.json` file, with epoch-level values read from matching `training_history.csv` logs.

IV. METHOD

Fig. 1 summarizes the paired multimodal pipeline. The image branch receives a Sentinel-2 RGB chip and extracts a spatial representation. The photon branch receives a compact ICESat-2 feature vector and maps it through an MLP with dropout. The branches are merged by concatenation, product fusion, fixed averaging, an original gating module, or an availability-aware gate. The fused representation is passed to a classifier head for three-class prediction.

Fig. 1. Paired multimodal remote-sensing fusion pipeline



V. EXPERIMENTAL SETUP

All five primary fusion variants were trained for 50 logged epochs. The reported held-out metrics are accuracy, macro-F1, balanced accuracy, Cohen kappa, and expected calibration error. Macro-F1 and balanced accuracy are emphasized because the confusion matrices show class imbalance; accuracy can remain high even when minority classes are weakly recovered.

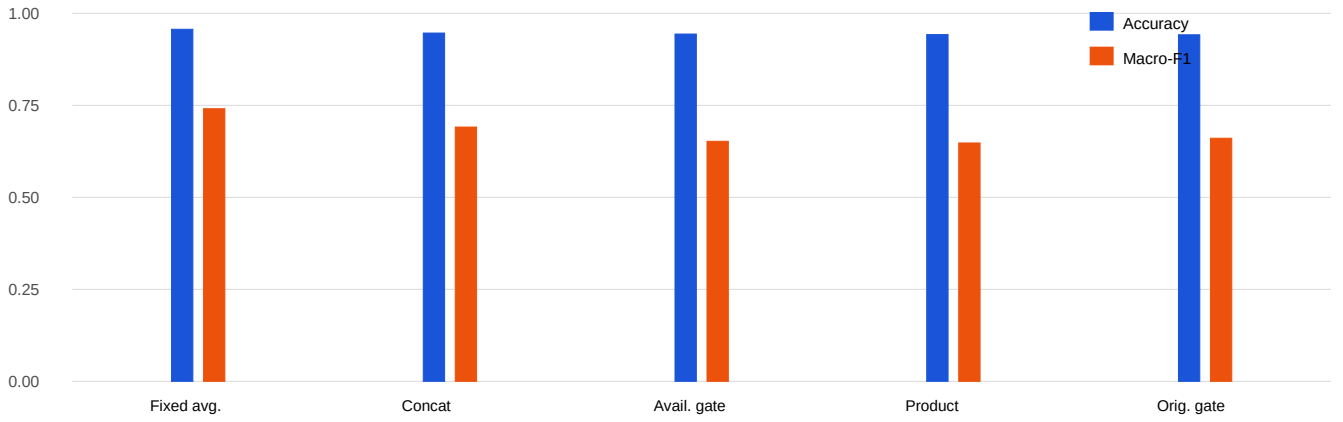
VI. RESULTS

Table I reports the primary held-out results. Fixed-average fusion is the strongest variant by both accuracy and macro-F1, achieving 0.9567 accuracy and 0.7408 macro-F1. The original gate and availability-aware gate show similar accuracy, but their macro-F1 and balanced accuracy differ, indicating that fusion choice changes class-wise predictions even when aggregate accuracy changes modestly.

TABLE I

Held-out results for completed real A6000 fusion runs.

Variant	Acc.	Macro-F1	Bal. Acc.	Kappa	ECE
Fixed average	0.9567	0.7408	0.7143	0.7272	0.0344
Concat	0.9464	0.6910	0.6842	0.0492	0.5620
Availability-aware gate	0.9437	0.6523	0.6399	0.6513	0.0519
Product	0.9422	0.6477	0.6291	0.6329	0.0544
Original gate	0.9417	0.6606	0.6621	0.6552	0.0549

Fig. 2. Held-out accuracy and macro-F1 by fusion operator**Fig. 3. Product-fusion confusion matrix**

Ref 0	0.99 (41509)	0.01 (290)	0.00 (27)
Ref 1	0.43 (733)	0.49 (829)	0.07 (124)
Ref 2	0.09 (219)	0.51 (1268)	0.40 (1005)
	Pred 0	Pred 1	Pred 2

VII. DISCUSSION

The results suggest that a simple fixed-average fusion operator can outperform more expressive fusion rules on this held-out split. One plausible interpretation is that averaging regularizes the multimodal decision boundary, while product and gated fusion introduce stronger interactions that may be sensitive to class imbalance or modality-specific noise. This interpretation should be tested further with additional seeds and geographic splits before broad claims are made. The concat result also deserves additional audit because its kappa is much lower than expected given its high accuracy and macro-F1; this inconsistency is reported explicitly rather than hidden.

VIII. CONCLUSION

This paper presents a reproducible paired Sentinel-2/ICESat-2 multimodal classification study designed for IEEE TGRS-style reporting. Five fusion operators were evaluated using real completed A6000 outputs. Fixed-average fusion currently gives the strongest held-out result, with 0.9567 accuracy and 0.7408 macro-F1. The main scientific takeaway is that fusion choice materially affects class-sensitive metrics, and that macro-F1, balanced accuracy, kappa, calibration, and confusion matrices should accompany accuracy in remote-sensing multimodal evaluation.

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